

LEVERAGE

1. Concept and Meaning of Leverage

Leverage is the use of fixed costs or fixed financial commitments to magnify the effect of a change in sales or operating profit on the earnings of a business. It works like a mechanical lever: a small movement in one variable can create a proportionately larger movement in another variable.

In financial management, leverage arises because some costs do not change with the level of activity. These fixed commitments may be operating costs, such as rent, depreciation and managerial salaries, or financial costs, such as interest on debt and preference dividend.

2. Main Types of Leverage

Type	Meaning
Operating Leverage	Arises from fixed operating costs and measures the sensitivity of EBIT to a change in sales.
Financial Leverage	Arises from fixed financial charges and measures the sensitivity of EPS to a change in EBIT.
Combined Leverage	Shows the joint effect of operating and financial leverage and measures the sensitivity of EPS to a change in sales.

3. Characteristics of Leverage

1. Leverage Arises Because of Fixed Costs

Leverage exists when a business has fixed costs. Fixed costs are expenses that do not change immediately with the level of production or sales. There are two main types of fixed costs:

- Fixed operating costs, such as rent, salaries, insurance and depreciation
- Fixed financial costs, such as interest on loans and preference dividends

Operating leverage arises because of fixed operating costs, while financial leverage arises because of fixed financial charges. Without fixed costs, the effect of leverage will be very low or may not exist.

2. Leverage Magnifies Profit

Leverage increases the effect of a change in sales or operating profit. A small percentage change in sales can cause a larger percentage change in EBIT or EPS. For example, if sales increase by 10% and EBIT increases by 20%, the business has experienced a magnifying effect. This happens because fixed costs remain unchanged. Once the fixed costs are covered, additional sales contribute more toward profit. Therefore, leverage works like a multiplier of profit.

3. Leverage Increases Risk

Higher leverage also increases the risk of the business. Fixed costs must be paid even when sales or profits are low. A company with high fixed operating costs faces greater business risk because expenses such as rent, salaries and depreciation continue even during a fall in sales. Similarly, a company with high debt faces financial risk because interest must be paid regularly, whether the company earns profit or not.

4. Leverage Can Increase Shareholders' Return

Leverage can increase the earnings available to equity shareholders when the business performs well. For example, if a company borrows money at 8% interest and earns a return of 14% from that money, the additional 6% benefit may increase the earnings of equity shareholders. In this situation, financial leverage works positively. However, if the company earns less than the cost of debt, shareholders' returns may decrease. Therefore, leverage increases returns only when the business earns enough profit to cover its fixed costs and financial charges.

5. Leverage Has a Two-Way Effect

Leverage affects both profit and loss. It can increase profit during favourable business conditions, but it can also increase loss during unfavourable conditions. When sales rise, a highly leveraged firm may experience a large increase in profit. But when sales fall, the same firm may experience a sharp decline in profit. For this reason, leverage is known as a **double-edged sword**. It provides an opportunity to earn higher returns, but at the same time it also creates the possibility of greater losses.

6. Leverage Helps in Decision-Making

Leverage is an important tool for management decisions. It helps managers select a suitable combination of fixed and variable costs. Operating leverage helps management decide whether the company should use more machinery, automation or labour. Financial leverage helps management decide whether the business should raise funds through:

- Equity shares
- Preference shares
- Debentures
- Bank loans
- Retained earnings

By studying leverage, management can choose a cost structure and capital structure that provide a proper balance between profit and risk.

7. Leverage Helps in Profit Planning

Leverage is useful for estimating future profits. It helps management understand how a change in sales may affect EBIT and EPS. Operating leverage measures the effect of sales changes on EBIT. Financial leverage measures the effect of EBIT changes on EPS. Combined leverage measures the total effect of a sales change on EPS. Therefore, leverage is useful for profit forecasting, budgeting, break-even analysis and future planning.

8. Degree of Leverage Changes with the Level of Activity

The degree of leverage does not remain the same at every level of sales and profit. It changes when sales, contribution, EBIT, fixed costs or interest charges change. Operating leverage is usually high when the company is operating near the break-even point. At this level, a small change in sales can cause a large change in EBIT. Financial leverage is high when EBIT is only slightly higher than interest charges. In this situation, a small change in EBIT can cause a large change in EPS. As sales and profits increase, the degree of leverage generally decreases because the company becomes more capable of covering its fixed costs. Therefore, leverage should always be measured at a specific level of sales and profit.

4. Basic Profit Relationships

$$\text{Contribution} = \text{Sales} - \text{Variable Cost}$$

$$\text{EBIT} = \text{Contribution} - \text{Fixed Operating Cost}$$

$$\text{EBT} = \text{EBIT} - \text{Interest}$$

$$\text{EPS} = \text{Earnings Available to Equity Shareholders} / \text{Number of Equity Shares}$$

OPERATING LEVERAGE

5. Meaning of Operating Leverage

Operating leverage refers to the use of fixed operating costs in the cost structure of a business. It measures how strongly Earnings Before Interest and Tax (EBIT) responds to a change in sales. It is created by costs such as factory rent, insurance, depreciation, permanent salaries and other fixed production or administrative expenses.

When fixed operating costs are high, each additional unit sold contributes more toward EBIT after variable costs are covered. However, when sales fall, the same fixed costs continue and EBIT declines more sharply.

6. Characteristics of Operating Leverage

- **Origin:** It arises because of fixed operating costs.
- **Risk measured:** It measures business risk arising from fluctuations in sales, demand, price and operating costs.
- **Profit effect:** A small percentage change in sales produces a larger percentage change in EBIT.
- **Independent of financing:** Interest, debt and capital structure do not affect operating leverage.
- **Industry relevance:** It is normally high in capital-intensive businesses such as manufacturing, airlines, telecom, hotels and automobiles.
- **Planning use:** It helps in cost control, automation decisions, sales planning and profit forecasting.

7. Measurement: Degree of Operating Leverage (DOL)

$$\text{At a given sales level} \\ \text{DOL} = \text{Contribution} / \text{EBIT}$$

$$\text{Between two sales levels} \\ \text{DOL} = \% \text{ Change in EBIT} / \% \text{ Change in Sales}$$

Interpretation: If DOL = 3, a 1% change in sales will cause approximately a 3% change in EBIT in the same direction, assuming the cost structure remains unchanged.

8. Factors Affecting Operating Leverage

- Amount of fixed operating cost
- Contribution margin per unit
- Sales volume and distance from the break-even point
- Degree of automation and capital intensity
- Selling price and variable cost structure

Risk rule: High fixed operating cost → High DOL → High business risk → Greater fluctuation in EBIT.

9. Effect of Operating Leverage on Profit

Situation	Effect on EBIT
Sales increase	EBIT rises faster than sales because fixed operating costs remain unchanged.
Sales decrease	EBIT falls faster than sales because fixed operating costs must still be paid.

FINANCIAL LEVERAGE

Concept, Characteristics and Measurement

10. Meaning of Financial Leverage

Financial leverage refers to the use of debt, debentures, preference shares or other sources carrying fixed financial charges. It measures how strongly Earnings Per Share (EPS) responds to a change in EBIT. Interest on debt must be paid irrespective of the level of operating profit, so debt can magnify the return available to equity shareholders as well as their risk.

Financial leverage is favourable when the return earned on borrowed funds is higher than their cost. It becomes unfavourable when the return on investment falls below the interest or other fixed financing cost.

11. Characteristics of Financial Leverage

- **Origin:** It arises from fixed financial charges such as interest and preference dividend.
- **Risk measured:** It measures financial risk, including the possibility of failing to meet fixed financial obligations.
- **Profit effect:** A small percentage change in EBIT produces a larger percentage change in EPS.
- **Capital structure based:** It depends on the proportion of debt and preference capital in the financing mix.
- **Independent of operating costs:** Financial leverage is determined after EBIT and is not directly affected by fixed operating costs.
- **Shareholder impact:** It can raise EPS and return on equity, but can also reduce them sharply when EBIT declines.

12. Measurement: Degree of Financial Leverage (DFL)

When only interest-bearing debt exists

$$DFL = EBIT / EBT$$

General sensitivity measure

$$DFL = \% \text{ Change in EPS} / \% \text{ Change in EBIT}$$

When preference shares exist

$$DFL = EBIT / [EBIT - \text{Interest} - \text{Preference Dividend} / (1 - \text{Tax Rate})]$$

Interpretation: If $DFL = 2$, a 1% change in EBIT will cause approximately a 2% change in EPS in the same direction.

13. Effect of Financial Leverage on Profit

Situation	Effect on EPS
EBIT increase	EPS increases faster because interest remains fixed.
EBIT decrease	EPS decreases faster because interest must still be paid.

LEVERAGE AND PROFIT ANALYSIS

14. Profitability Statement for Leverage Analysis

Particulars	Relationship
Sales Revenue	Total revenue from sales
Less: Variable Cost	Changes with output or sales
Contribution	Sales - Variable Cost
Less: Fixed Operating Cost	Remains fixed within relevant range
EBIT	Contribution - Fixed Operating Cost
Less: Interest	Fixed financial charge
EBT	EBIT - Interest
Less: Tax	Tax on EBT
Less: Preference Dividend	Fixed claim after tax
Earnings for Equity	Residual earnings
EPS	Earnings for Equity / No. of shares

15. Separate Effects on Profit

Leverage	Effect on Profit
Operating leverage	Converts a change in sales into a larger change in EBIT. Its strength depends on contribution and fixed operating cost.
Financial leverage	Converts a change in EBIT into a larger change in EPS. Its strength depends on interest, preference dividend and capital structure.
Combined leverage	Converts a change in sales into a larger change in EPS by combining both operating and financing effects.

16. Favourable and Unfavourable Effects

- **Favourable operating effect:** Sales grow above the break-even level, so contribution increases while fixed operating cost remains unchanged.
- **Unfavourable operating effect:** Sales fall, but fixed operating costs continue, causing a sharp decline in EBIT.
- **Favourable financial effect:** EBIT and return on investment are sufficient to cover interest, allowing equity shareholders to receive a higher EPS.
- **Unfavourable financial effect:** EBIT declines or the return on borrowed funds is below the financing cost, reducing EPS and increasing default risk.

COMBINED LEVERAGE

18. Meaning of Combined Leverage

Combined leverage, also called total or composite leverage, represents the joint effect of operating leverage and financial leverage. It measures the sensitivity of EPS to a change in sales. Fixed operating costs first magnify the effect of sales on EBIT, and fixed financial charges then magnify the effect of EBIT on EPS.

19. Measurement: Degree of Combined Leverage (DCL)

$$\text{DCL} = \text{DOL} \times \text{DFL}$$

When only interest-bearing debt exists

$$\text{DCL} = \text{Contribution} / \text{EBT}$$

Sensitivity measure

$$\text{DCL} = \% \text{ Change in EPS} / \% \text{ Change in Sales}$$

Interpretation: If DCL = 6, a 1% change in sales will cause approximately a 6% change in EPS in the same direction.

20. Characteristics of Combined Leverage

- **Total-risk measure:** It combines business risk and financial risk.
- **Sales-to-EPS link:** It shows the final effect of a sales change on equity shareholders.
- **Multiplicative relationship:** A high DOL combined with a high DFL produces a very high DCL.
- **Risk balancing:** A firm with high operating leverage should generally use lower financial leverage, and vice versa.

21. Difference Between Operating Leverage and Financial Leverage

• Basis	• Operating Leverage	• Financial Leverage
• Meaning	• Operating leverage shows the effect of a change in sales on operating profit or EBIT.	• Financial leverage shows the effect of a change in EBIT on EPS or earnings available to equity shareholders.
• Cause	• It arises because of fixed operating costs.	• It arises because of fixed financial charges.
• Fixed costs involved	• Rent, depreciation, insurance, fixed salaries and machinery costs.	• Interest on loans, debentures and preference dividends.
• Profit affected	• It affects Earnings Before Interest and Tax, or EBIT.	• It affects Earnings Before Tax, net profit and Earnings Per Share, or EPS.
• Relationship measured	• It measures the relationship between sales and EBIT.	• It measures the relationship between EBIT and EPS.
• Formula	• $\text{DOL} = \text{Contribution} \div \text{EBIT}$	• $\text{DFL} = \text{EBIT} \div \text{EBT}$
• Percentage formula	• $\text{DOL} = \% \text{ Change in EBIT} \div \% \text{ Change in Sales}$	• $\text{DFL} = \% \text{ Change in EPS} \div \% \text{ Change in EBIT}$
• Type of risk	• It measures business risk.	• It measures financial risk.
• Main decision	• It helps management choose the proper combination of fixed and variable operating costs.	• It helps management choose the proper debt-equity mix.
• Source of risk	• Risk arises because fixed operating costs must be paid even when sales are low.	• Risk arises because interest and other financial charges must be paid even when EBIT is low.

• Favourable effect	• When sales increase, EBIT increases at a faster rate.	• When EBIT increases, EPS increases at a faster rate.
• Unfavourable effect	• When sales decrease, EBIT decreases at a faster rate.	• When EBIT decreases, EPS decreases at a faster rate.
• Break-even point	• It is highest near the operating break-even point, where contribution is close to fixed operating costs.	• It is highest near the financial break-even point, where EBIT is close to interest charges.
• Relation with financing	• It is independent of the financing method used by the company.	• It depends on the amount of debt and preference capital used by the company.
• Used by	• It is mainly related to production, sales and cost structure decisions.	• It is mainly related to financing and capital structure decisions.